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horizontal tiers, with the end of the cells of one tier appearing at approximately the same level, as seen in tangential longitudinal section (TLS), it is called **storied (stratified) cambium**. It is the characteristic of the plants with short fusiform initials. Whereas in plants with long fusiform initials, they strongly overlap at the ends, and this type of cambium is called **non-storied (non-startified) cambium**.

Ray Initials

These are horizontally elongated cells. They give rise to the ray cells and form the elements of the radial system of secondary xylem and phloem.

Activity of Vascular Cambium

The vascular cambial ring, when active, cuts off new cells both towards the inner and outer side. The cells which are produced outward form secondary phloem and inward secondary xylem.

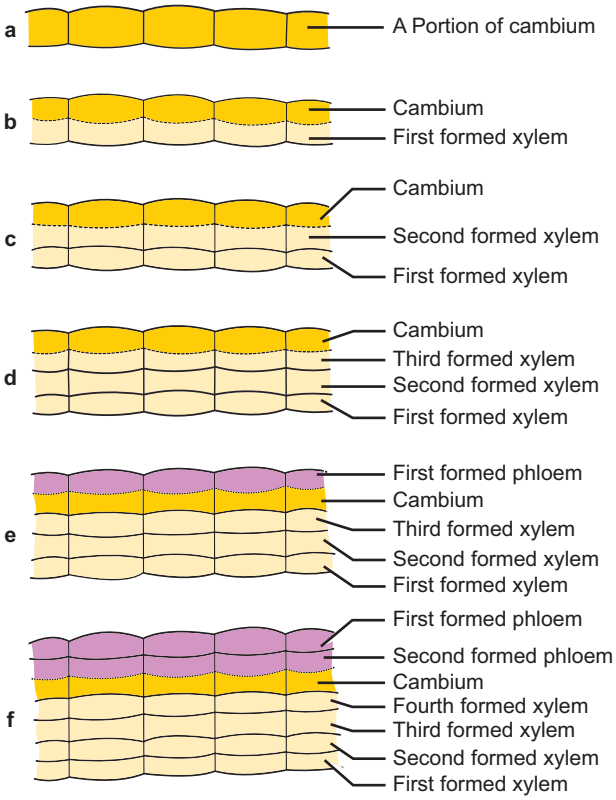


Figure 10.3: Diagrammatic representation of vascular cambial activity (a–f)

At places, cambium forms some narrow horizontal bands of parenchyma which passes through secondary phloem and xylem. These are the rays.

Due to the continued formation of secondary xylem and phloem through vascular cambial activity, both the primary xylem and phloem get gradually crushed.

Secondary Xylem

The secondary xylem, also called **wood**, is formed by a relatively complex meristem, the vascular cambium, consisting of vertically (axial) elongated fusiform initials and horizontally (radially) elongated ray initials.



Xylotomy

The study of wood by preparing sections for microscopic observation.

The axial system consists of vertical files of treachery elements, fibers, and wood parenchyma. Whereas the radial system consists of rows of parenchymatous cells oriented at right angles to the longitudinal axis of xylem elements.

The secondary xylem varies very greatly from species to species with reference to relative distribution of the different cell types, density and other properties. It is of two types.

Porous Wood or Hard Wood

Generally, the dicotyledonous wood, which has vessels is called **porous wood** or **hard wood**. Example: *Morus rubra.*

Non- Porous Wood or Soft Wood